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Chapter 33

GELFAND-P001491

3/X̄-88. Gelfand Seminar.

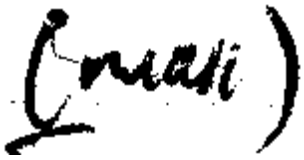


Uncertain reading. In the digital image a yellow icon sits to the left of the digit 3. The area beneath it is obscured.

Original source details: GELFAND-P001491

What should one read in mathematics? (for first-year students)

Beilinson. (outline)



Uncertain reading. The short entry in parentheses tentatively reads «план» (“outline”).

Original source details: GELFAND-P001491

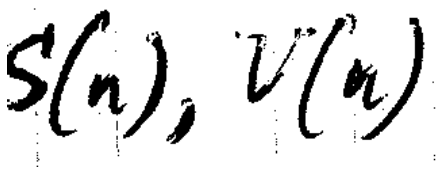
1. Linear algebra.
2. Functions of a complex variable.
3. Galois theory.
4. Elementary algebraic topology.
5. Representation theory.
6. Algebraic geometry (elliptic curves, etc.).
7. Arithmetic.

GELFAND-P001492

[source number: 4]

Kirillov (outline)

1. Linear algebra
2. Analysis — real, complex and p-adic
3. Operator theory + quantum mechanics.
4. Harmonic analysis + integral geometry
5. Representations of $S(n)$, $V(n)$ and combinatorics.



Uncertain reading. The second letter is shaped like V. U is also possible.

Original source details: GELFAND-P001492

6. Elements of algebraic geometry (commutative and noncommutative).
7. Classical works (a selection).

Vasiliev Vitya

1. Linear [crossed out] algebra.

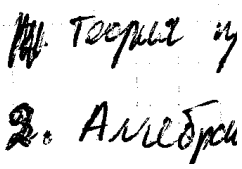


Original source details: GELFAND-P001492

Uncertain reading. The crossed-out fragment before “algebra” resembles «алг.» (“alg.”).

[number crossed out] Representation theory.

2. Algebraic and differential topology, homological algebra.



Original source details: GELFAND-P001492

Uncertain reading. The number before “Representation theory” is crossed out. The following number is overwritten and reads 2.

3. Dynamical systems, ergodic theory, mechanics.
4. Algebraic geometry (geometric chapters).
5. Functional analysis.

Goncharov Sasha

1. Linear algebra.
2. Theory of functions of a complex variable [ТФКП].
3. Analysis (generalized functions ...)
4. Algebraic topology, homological algebra, algebraic geometry (classical)
5. Algebraic geometry (modern)
6. Differential geometry.
7. Representation theory (finite-dimensional and ∞-dimensional)
8. Arithmetic
   a) classical
   b) class field theory
9. F. Klein, “Lectures on the Development of Mathematics in the XIX Century”

GELFAND-P001493

Kapranov Misha

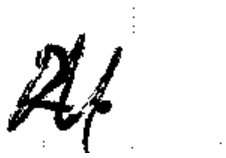
[At upper right: (here linear algebra [crossed out] is omitted as uninformative)]



Original source details: GELFAND-P001493

Uncertain reading. The word after “algebra” is crossed out and illegible.

1. Elementary topology.
   (π_1, surfaces)



Original source details: GELFAND-P001493

Uncertain reading. The crossed-out number at left may be 2.

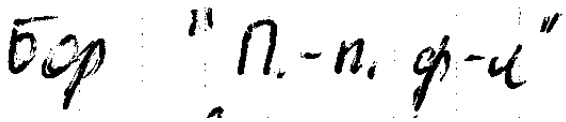
2. Galois theory.
3. Algebraic geometry.
4. Algebraic topology
5. Physics
6. Arithmetic.

Zelevinsky Andrei.

1. Algebra: Galois theory.
   etc.

I. M. Gelfand

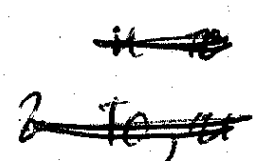
1. Feynman for schoolchildren
2. Bohr, “П.-п. ф-и” [abbreviated title]



Original source details: GELFAND-P001493

Uncertain reading. The surname reads «Бор» (Bohr), possibly «Бэр» (Ber).

3. Minkowski, “Geometrie der Zahlen”
4. Hurwitz, “Elliptic Functions”
5. Gelfand, “Linear Algebra” (chapters 1 and 2)
6. Mechanics, “Landau and Lifshitz”,
   calculus of variations, symplectic geometry,
   (especially Gromov)
7. Quantum mechanics



Original source details: GELFAND-P001493

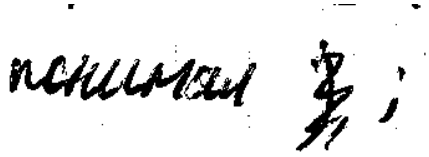
Uncertain reading. Two crossed-out lines. The first appears to start with “and …” (и); the second with «2. Те…».

8. Spectral theory of operators.

9. Group representation theory

(Serre, “Theory of Representations of Finite Groups” — only chapters 1 and 2!

[Below at right:] in the third he understood nothing [crossed out];

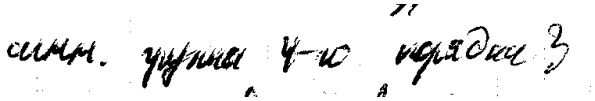


Uncertain reading. The crossed-out mark after “understood” is illegible.

Original source details: GELFAND-P001493

[Below at left:] Gelfand, Minlos, Shapiro)

Why is the group of motions of the cube the symmetric group of order 4?



Uncertain reading. The manuscript reads «4-го порядка» (of order 4).

Original source details: GELFAND-P001493

There are 2 equally valid geometries, Euclidean and symplectic,

GELFAND-P001494

[source number: 5]

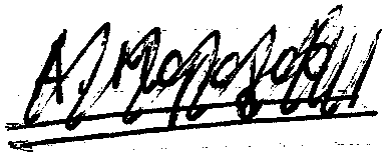
10. Statistical mechanics, chaos, ТФДП [source abbreviation], ergodic theory

11. p-adic analysis, strings, etc.

(especially, Morozov’s lectures)

12. Quantization in symplectic geometry, theory of D-modules

A. Yu. Morozov [oblique strokes]



Uncertain reading. The heading is double-underlined and crossed by oblique strokes.

Original source details: GELFAND-P001494

I. M. Gelfand

What is a particle’s spin?

[Inserted above: in the theory of relativity]

Consider the mass hyperboloid:

$$-p_1^2 - p_2^2 - p_3^2 + p_4^2 = c^2$$

[At right: (with [abbrev.] rest mass)]

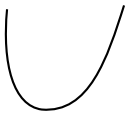


Uncertain reading. The abbreviation before “rest mass” is unclear. Possibly «нул.» (“zero”).

Original source details: GELFAND-P001494



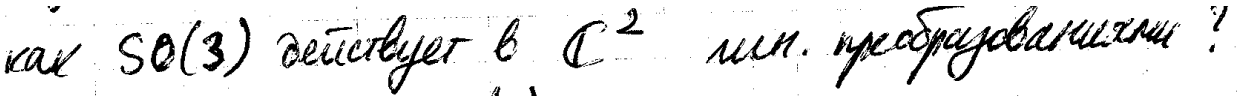
Diagram and typed description.



Original source details: GELFAND-P001494

At each point of the hyperboloid one must put a linear space characterizing its degree of freedom.

Exercise: how does $SO(3)$ act on $\mathbb{C}^2$ by linear transformations?



Original source details: GELFAND-P001494

Uncertain reading. The manuscript reads $SO(3)$ and $\mathbb{C}^2$.

(see Gelfand–Minlos–Shapiro, chapter 1)

Problem. Construct the mathematics corresponding to the case of arbitrary spin

(spin 0 — ordinary; spin $1/2$ — super).

Fröhlich explains how the braid group is applied.

A. Morozov.

In 3-dimensional space there is linking of contours.

The Bohm–Aharonov effect

(Bohm-Aharonov)

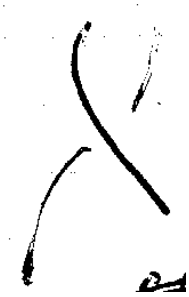


Diagram and typed description.



Original source details: GELFAND-P001494

[crossed out]



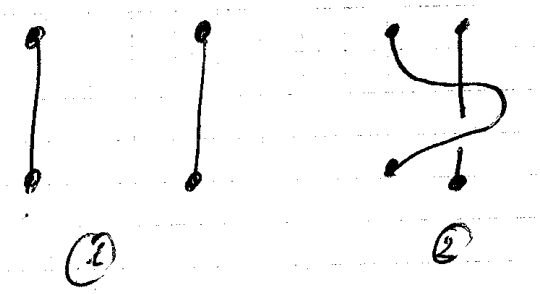
Undeciphered passage. The word above "Spin?" is illegible.

Original source details: GELFAND-P001494

Spin?

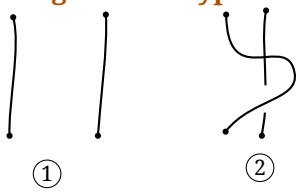
spin θ

(in the future)



Original source details: GELFAND-P001494

Diagram and typed description.



worldlines

$\mathbb{R}^1_t \times \mathbb{R}^2$

[At right: (in $\mathbb{R}^3$ and only there is there a fundamental difference between ① and ②)]

	①	②
Amplitude	1	$e^{2\pi i\theta}$

[At right: (earlier θ — spin)]

This applies to the quantum Hall effect.

On [taking it?] downward, some number of circuits arises.



Original source details: GELFAND-P001494

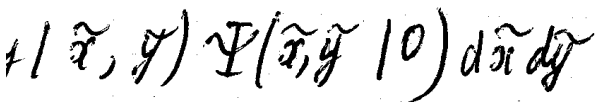
Uncertain reading. The word before “downward” is tentatively read «унесении» (taking away).

GELFAND-P001495

What happens to the wave function

$$\begin{aligned} d > 3 &\implies \theta = \frac{\text{integer}}{2} \\ d = 3 &\implies \theta \text{ — arbitrary!} \end{aligned}$$

$$\Psi(x, y; T) = \int A_T(x, y \mid \tilde{x}, y)\, \Psi(\tilde{x}, \tilde{y} \mid 0)\, d\tilde{x}\, d\tilde{y}$$



Original source details: GELFAND-P001495

Uncertain reading. In A_T the second argument after the bar has no distinct tilde. The top stroke joins the y.

[Crossed out beneath the integral sign]



Original source details: GELFAND-P001495

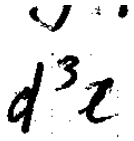
Uncertain reading. Strokes resembling d and y are visible beneath the cancellation.

$$A_T = \sum_{\substack{\text{over particle} \\ \text{trajectories}}} e^{L_1} e^{L_2} \underbrace{F(\text{trajectories})}_{\substack{\| \\ e^{2\pi i\theta\,(\text{index})}}} =$$

(how many times they became linked)

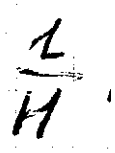
θ half-integer except [inserted: the case] $d = 3$

$$= \int \mathcal{D}\varphi\, \mathcal{D}A \quad \exp \int_{d^3 z} |D\varphi|^2 + \frac{1}{H} F_{\mu\nu}^2 + \epsilon_{\mu\nu\lambda} A_\mu F_{\nu\lambda}$$



Uncertain reading. The letter after d^3 looks like z. Possibly x.

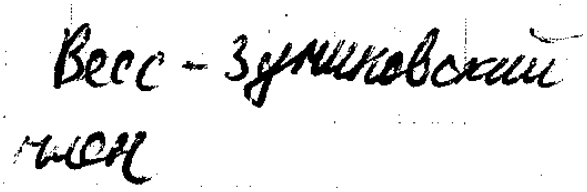
Original source details: GELFAND-P001495



Uncertain reading. The denominator looks like H.

Original source details: GELFAND-P001495

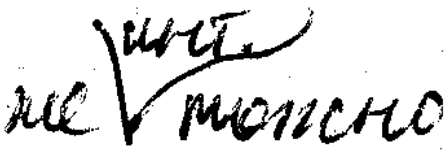
[Beneath the last summand:] Wess–Zumino term



Uncertain reading. The last summand is labelled «Весс-Зуминовский член» (Wess–Zumino term).

Original source details: GELFAND-P001495

The same [inserted: integrals [abbrev.]] can be used for high-temperature conductivity [source abbreviation: провод.].



Uncertain reading. The insertion appears to read «инт.».

Original source details: GELFAND-P001495

Witten recently used it to describe knot invariants

(Jones polynomials)

S. Shatashvili (LOMI)

P. Wiegmann

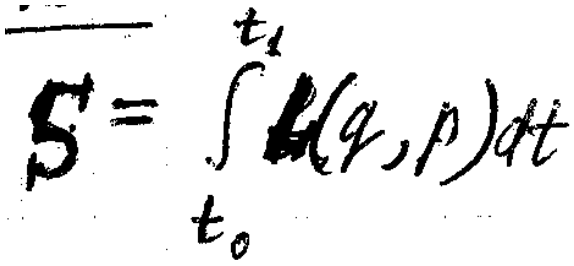
GELFAND-P001496

17/X-88. Gelfand seminar.

I. M. Gelfand. 1) Hierarchy of Functionals

Local functionals:

$$S = \int_{t_0}^{t_1} H(q,p)\,dt$$



Uncertain reading. The letter before (q,p) is overwritten. It is read tentatively as H, possibly L.

Original source details: GELFAND-P001496

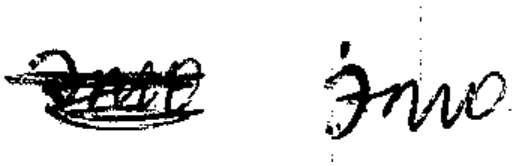
Functionals of the next level: if the variation δS is local.

(of the 1st kind)

[At right: || Example: λ_1]

Functionals of the 2nd kind: if δS — of the 1st kind.

~~This~~ This is a new subject of the homological-algebra type.



Uncertain reading. The crossed-out word before “This is a new subject” appears to be a repeated “This”.

Original source details: GELFAND-P001496

2) Quantum mechanics: amplitude

$$e^{\frac{i}{\hbar}\int_{t_0}^{t_1} L(p,q)\,dt}$$

Read Einstein and Bohr’s discussion of the uncertainty principle.

One must write

$$\int e^{\frac{i}{\hbar}\int_{t_0}^{t_1} L(p,q)\,dt}\,Dp\,Dq$$

The stationary-phase method.

GELFAND-P001497

Problem: calculate approximately

$$\int_0^1 (\sin 10^5 x^3)\, dx$$

An engaging problem: a dual exposition of quantum mechanics, through which one can understand much mathematics

One should read: a) Landau, Lifshitz, “Quantum Mechanics”

Feigin B. Strings.

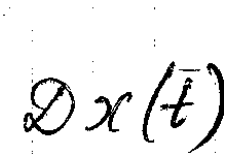
Quantum mechanics:

M is a manifold, to every map of the circle $S^1 \to M$ one assigns

$$e^{\frac{i}{\hbar}\int \frac{m\dot{x}^2}{2}\, dt} = e^{\frac{i}{\hbar}\int \mathcal{L}(x,\dot{x})\, dt} \tag{1}$$

and one must write

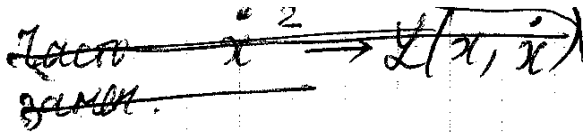
$$\int e^{\frac{i}{\hbar}\int \frac{m\dot{x}^2}{2}\, dt} \mathcal{D}x(\bar{t}) = \int e^{\frac{i}{\hbar}\int \mathcal{L}(x,\dot{x})\, dt} \mathcal{D}x(t) \tag{1}'$$



Uncertain reading. A separate bar is visible over t in the first $\mathcal{D}x(\bar{t})$. The second $\mathcal{D}x(t)$ has no such bar.

Original source details: GELFAND-P001497

[Crossed out: Част. / Чисто; замен. [source wording] $\dot{x}^2 \longrightarrow \mathcal{L}(x, \dot{x})$.]

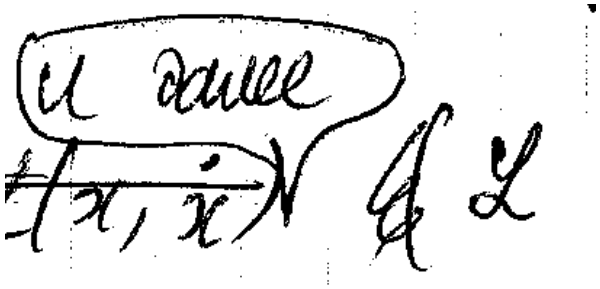


Uncertain reading. The first crossed-out word looks like «Част.» or «Чисто». The second looks like «замен.» (replace).

Original source details: GELFAND-P001497

[Inserted and circled: And even]

& $\mathcal{L}$ must be taken more general.



Uncertain reading. The sign before $\mathcal{L}$ resembles &. It may be a rewritten conjunction.

Original source details: GELFAND-P001497

Let M be a symplectic manifold

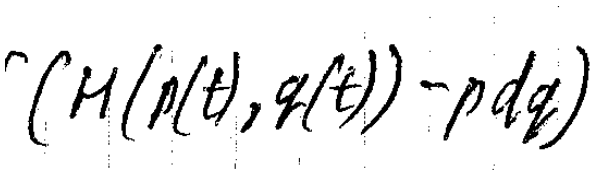
p, q are coordinates, H is a function on M

Write the action for a trajectory $p(t), q(t)$

$$\int [H(p(t), q(t))\, dt - p\, dq] \tag{2}$$

and

$$\int e^{\frac{i}{\hbar}\int (H(p(t),q(t)) - p\, dq)}\, Dp\, Dq \tag{2'}$$



Uncertain reading. Equation (2′) gives no dt after $H(p(t), q(t))$. Equation (2) has dt there.

Original source details: GELFAND-P001497

The approaches using (1′) and (2′) are related by the Legendre transformation

~~Now one must~~

A more invariant expression: $S^1 \to M$

$$\int e^{\frac{i}{\hbar}\left[\int_{S^1} H\, dt - \int_{\oslash} \omega\right]} \tag{3}$$



Diagram and typed description. ⊘

Original source details: GELFAND-P001497

[At right:] (ω is the symplectic form,

⊘ — [Crossed out: disk?; inserted: membrane], spanning S^1

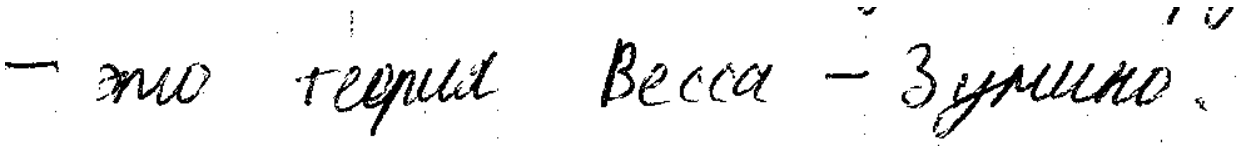


Uncertain reading. The crossed-out word beneath “membrane” may be “disk”.

Original source details: GELFAND-P001497

The generalization of (3) to strings in a certain special case —

— is the Wess–Zumino theory.



Original source details: GELFAND-P001497

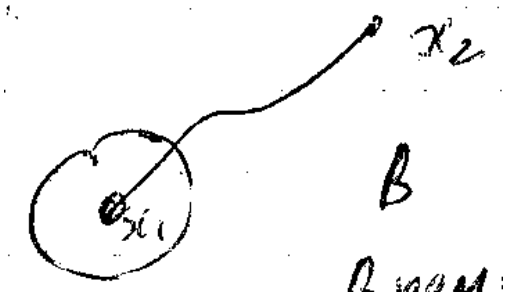
Uncertain reading. The word between «это» and «Весса» reads «теория» (theory).

GELFAND-P001498

[source number: 41]

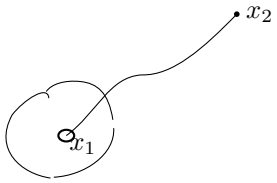
I. M. G. Legendre transformation: see Gelfand–Fomin's book on the calculus of variations (appendix)

(via wavefronts)



Original source details: GELFAND-P001498

Diagram and typed description.



The outline has gaps and a short hooked stroke. The purpose of the stroke is unclear.

A "wind rose" is specified at every point

Time of motion

$$\int_{x_1}^{x_2} L(x, \dot{x})\, dt$$

(independent of parametrization)

Take wavefronts.

Trajectories are characteristics.

— · —

F. $\mathcal{E}$ is a Riemannian 2-dimensional manifold; g_{ij} is the metric tensor on $\mathcal{E}$.

Free Lagrangian (analogue of $\dot{x}^2$)

$$\mathcal{L}(X) = \int_{\mathcal{E}} \sqrt{g}\, g^{\alpha\beta} \frac{\partial X^a}{\partial x^\alpha} \frac{\partial X^b}{\partial x^\beta} G_{ab}\, dx^0\, dx^1 \tag{4}$$



Original source details: GELFAND-P001498

Uncertain reading. Equation (4) shows G_{ab} and a cancelled superscript. The following line uses G^{ab}.



Original source details: GELFAND-P001498

Uncertain reading. The first numeral in $dx^0\, dx^1$ is heavily retraced and looks like 0. The line below gives x^1, x^2.

$\{X^a(x^1, x^2)\}$ is an embedding of $\mathcal{E}$ into $\mathbb{R}^d$; G^{ab} is the metric in $\mathbb{R}^d$

Feynman integral

[Crossed out at left]

$$\int e^{\frac{i}{\hbar}\mathcal{L}(X)}\,\mathcal{D}X\,\mathcal{D}g \tag{4'}$$



Uncertain reading. The crossed-out label before (4′) resembles (4).

Original source details: GELFAND-P001498

I. M. G.'s question: what is the Legendre transformation associated with surfaces rather than curves?

This is a fundamental and substantial question

Wess–Zumino theory.

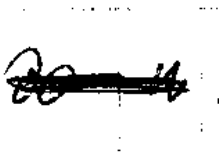
$\mathcal{E}$ with a complex structure. Write in local coordinates:

$z, \bar{z}$ are coordinates

$$\left.\begin{aligned}\frac{\partial}{\partial z} + A_0(z,\bar{z})\\ \frac{\partial}{\partial \bar{z}} + A_1(z,\bar{z})\end{aligned}\right\} \quad \text{connection} \tag{5}$$

Choose a 3-dimensional membrane whose boundary is $\mathcal{E}$

Complete [crossed out] the components (5) to flat connections

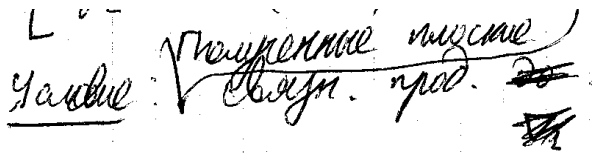


Uncertain reading. After "Complete," «до» ("to") and one further stroke are crossed out.

Original source details: GELFAND-P001498

$$\begin{aligned}\left[\frac{\partial}{\partial \bar{z}} + B_0(z,\bar{z}),\ \frac{\partial}{\partial z} + A_0(z,\bar{z})\right] = 0\\ \left[\frac{\partial}{\partial z} + B_1(z,\bar{z}),\ \frac{\partial}{\partial \bar{z}} + A_1(z,\bar{z})\right] = 0\end{aligned} \tag{6}$$

(7) Condition: [inserted above: The resulting flat] conn. extend [crossed out] onto the membrane as flat connections ∇_1, ∇_2



Uncertain reading. After «прод.» ("extend"), «до» ("to") is crossed out. The cancelled symbol below it resembles ∇_1.

Original source details: GELFAND-P001498

GELFAND-P001499

Lagrangian:

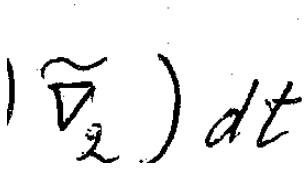
$$\mathcal{L}(A_0, A_1) = \int (A_0 - B_1)(B_0 - A_1)\, dz \wedge d\bar{z} + \underset{\text{(Wess-Zumino)}}{WZ}(\widetilde{\nabla}_1, \widetilde{\nabla}_2), \tag{8}$$

The notation used above is

$$\nabla_1 = \left\{ \frac{\partial}{\partial z} + A_0,\ \frac{\partial}{\partial \bar{z}} + B_0 \right\},$$
$$\nabla_2 = \left\{ \frac{\partial}{\partial z} + B_1,\ \frac{\partial}{\partial \bar{z}} + A_1 \right\};$$

$\widetilde{\nabla}_j$ are flat extensions of ∇_j to the membrane

$$WZ(\widetilde{\nabla}_1, \widetilde{\nabla}_2) = \int_0^1 \int_{\text{membrane}} \operatorname{tr}((\widetilde{\nabla}_1 - \widetilde{\nabla}_2) \cdot \Phi(t\widetilde{\nabla}_1 + (1-t)\widetilde{\nabla}_2)\, dt \tag{9}$$



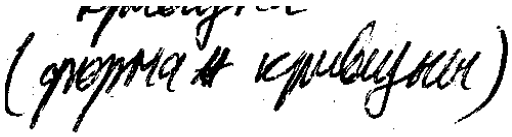
Original source details: GELFAND-P001499

Uncertain reading. Two opening parentheses follow tr. Before dt only the parenthesis closing the argument of Φ is visible.

[A stroke from Φ leads to the caption]

curvature

(form [crossed out] of curvature)



Original source details: GELFAND-P001499

Uncertain reading. A short mark between “form” and “of curvature” is crossed out.

I. M. Gelfand

Determinant of a cubic matrix.

(after [crossed out] I. M. G., Zelevinsky and Kapranov)



Original source details: GELFAND-P001499

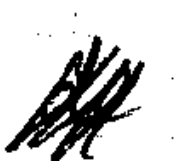
Uncertain reading. The crossed-out surname may start «Зелевин…» (Zelevin…).

introduction to a future talk

$$a_{i,j,k}, \quad i,j,k = 1,2.$$

$$\det \|\lambda a_{ij1} + \mu a_{ij2}\|$$

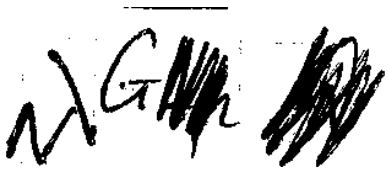
a quadratic form in λ and μ.
Take its discriminant in λ, μ. It is called the determinant of the cubic matrix.
In det A there is the action XAY,
and in the quadratic form X^*AX.
The group action is important.
The whole of group theory arose from the concept of det.
From cubic [crossed out] matrices new objects should arise.



Original source details: GELFAND-P001499

Undeciphered passage. The fragment before "matrices" is illegible.

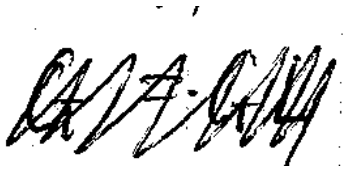
Representations of GL_n or SL_n.
N — a unipotent subgroup; ${}_N\backslash G$ [crossed out] — the space of affine flags



Original source details: GELFAND-P001499

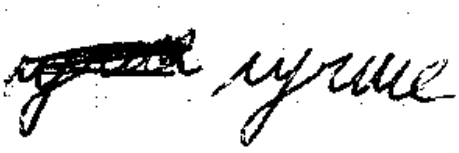
Undeciphered passage. Two groups of signs after ${}_N\backslash G$ are crossed out.

(look in detail at GL_2); [crossed out] $G = SL_2$ [crossed out] is better



Original source details: GELFAND-P001499

Uncertain reading. The crossed-out notation before $G = SL_2$ contains shapes resembling G and A.



Original source details: GELFAND-P001499

Undeciphered passage. The crossed-out word before "better" is illegible.

On ${}_N\backslash G$ consider functions (polynomials), homogeneous under left shifts.

GELFAND-P001500

[source number: 42]

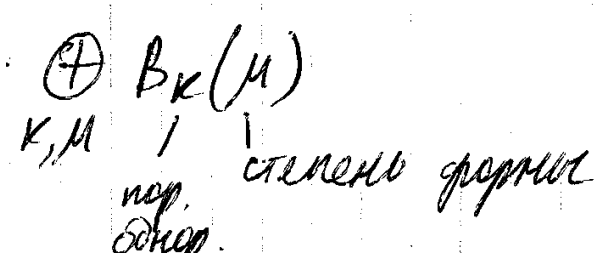
Example: the affine plane and polynomials of degree n

But there are also differential forms. Denote by B the space of forms ω,

such that $i_\lambda \omega = 0$ (λ is an element of the Lie algebra)

Now one must take $B + dB$

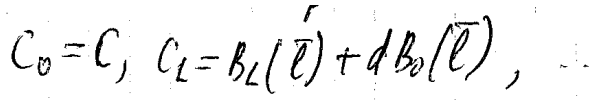
$$B = \bigoplus_{k,\mu} B_k(\mu)$$



Uncertain reading. A stroke from k points to "order of homogeneity" and one from μ to "degree of the form".

Original source details: GELFAND-P001500

$$C_0 = C, \quad C_1 = B_1(\bar{\ell}) + dB_0(\bar{\ell}), \ \ldots$$



Uncertain reading. The arguments in the formula for C_1 read as $\bar{\ell}$.

Original source details: GELFAND-P001500

Whitehead torsion arises: hyperdeterminant.

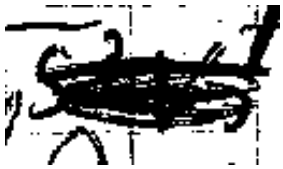
Apparently this makes it possible to replace Young diagrams.

GELFAND-P001501

24.10.88. Gelfand Seminar

I. M. G. Problems on representations.

There are representations, [crossed out] the trivial one; the tautological one in V; in S^2V (homogeneous polynomials of degree 2).



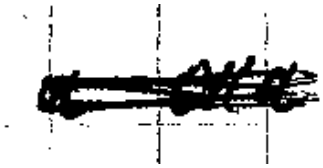
Undeciphered passage. The entry before "the trivial one" is crossed out and illegible.

Original source details: GELFAND-P001501

Prove irreducibility of $SL(2,\mathbb{R})$ in S^3V (polynomials of degree 3).

$\dim V = 2$, in V the tautological representation acts.

What about $SL(3,\mathbb{R})$? This is an important problem [crossed out] and it needs to be thought through.



Undeciphered passage. The entry after "problem" is crossed out. Its strokes are illegible.

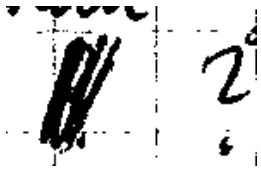
Original source details: GELFAND-P001501



Original source details: GELFAND-P001501

Uncertain reading. The last word of the line is read «продумать» (think through).

Hypothesis: homogeneous polynomials here too [crossed out]?



Original source details: GELFAND-P001501

Undeciphered passage. A short mark is heavily crossed out before the question mark.

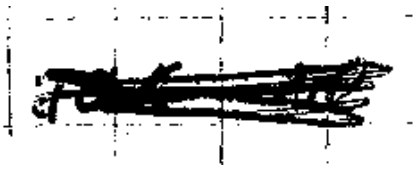
Gelfand, Naimark

3 important problems.

1) S_n (symmetric group)

2) $SL_n(\mathbb{R})$ — finite-dimensional representations

3) SO_n [crossed out] in the same detail as [crossed out] in the book by G.–M.–S., [below:] for SO_3.



Original source details: GELFAND-P001501

Undeciphered passage. An illegible fragment after SO_n is crossed out.



Original source details: GELFAND-P001501

Uncertain reading. The crossed-out entry after “as” may begin with “for”.



Original source details: GELFAND-P001501

Uncertain reading. A small detached mark appears below the line. Its status is unclear.

Representations of $SL(3,\mathbb{R})$.

The answer is obtained in Gelfand–Naimark’s book “Unitary Representations of the Classical Groups”.

N, B — subgroups in $G = SL(3,\mathbb{R})$.

GELFAND-P001502

[source number: 46]

$$\left\{ \begin{pmatrix} 1 & 0 & 0 \\ a_{21} & 1 & 0 \\ a_{31} & a_{32} & 1 \end{pmatrix} \right\} = N \tag{1}$$

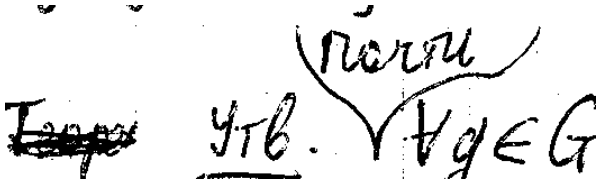
$$\left\{ \begin{pmatrix} b_{11} & 0 & 0 \\ b_{21} & b_{22} & 0 \\ b_{31} & b_{32} & b_{33} \end{pmatrix} \right\} = B \quad \begin{matrix} \text{(Borel)} \\ b_{11}b_{22}b_{33} = 1. \end{matrix} \tag{2}$$

Consider the cosets

$$\begin{matrix} N\backslash G \\ \| \\ \{Ng \mid g \in G\} \end{matrix} \quad \text{and} \quad B\backslash G \tag{3}$$

(4) ~~Thm.~~ Claim. [Inserted above ∀: Almost]

$$\forall g \in G,\ \exists \text{ a decomposition } g = nz, \quad \text{where } n \in N, \qquad z = \begin{pmatrix} z_{11} & z_{12} & z_{13} \\ 0 & z_{22} & z_{23} \\ 0 & 0 & z_{33} \end{pmatrix}.$$



Original source details: GELFAND-P001502

Uncertain reading. "Thm." before "Claim" is crossed out. "Almost" is inserted above ∀.

(denote by Z the group of such matrices)

(Gauss decomposition) (upper triangular)

The decomposition of any matrix generalizing this is called the Bruhat decomposition.

[crossed out] We will use matrices z as representatives in $N\backslash G$.



Original source details: GELFAND-P001502

Undeciphered passage. The initial mark before "We will" is crossed out and illegible.

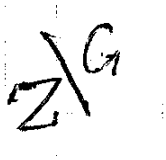
Define functions $f(z)$ [crossed out] on $N\backslash G$.



Original source details: GELFAND-P001502

Undeciphered passage. The mark after $f(z)$ is crossed out and illegible.

$Z\backslash G$ — the space of flags. (compactify further)

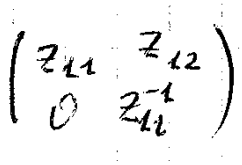


Uncertain reading. The manuscript here reads $Z\backslash G$.

Original source details: GELFAND-P001502

$$\begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ \alpha a + c & \alpha b + d \end{pmatrix}$$

$$\begin{pmatrix} z_{11} & z_{12} \\ 0 & z_{11}^{-1} \end{pmatrix}$$



Uncertain reading. The lower-right subscript reads 11. The final digit differs from the 2 above it.

Original source details: GELFAND-P001502

Construct the polynomial $f(z_{11}, z_{12})$

For order 3 one must take polynomials in the first two rows, homogeneous in each. They should be considered on the appropriate compactification.



Uncertain reading. The word before "compactification" reads «нужной» (appropriate).

Original source details: GELFAND-P001502

$$\lambda = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}, \tag{5}$$

multiplication [inserted: on the left] by λ takes a coset to a coset:

$$g_1 = ng_2 \;\Rightarrow\; \lambda g_1 = \lambda n g_2 = \underbrace{\lambda n \lambda^{-1}}_{n_1} \lambda g_2, \quad \text{as required.}$$



Uncertain reading. Two short fragments near the implication arrow are crossed out.

Original source details: GELFAND-P001502

GELFAND-P001503

General definition.

$Z = N\backslash G$ — the basic affine space.

Consider on Z all analytic functions. These are polynomials. Consider functions homogeneous under left translations:

$$f(\lambda z) = \lambda_1^{k_1}\lambda_2^{k_2}\lambda_3^{k_3} f(z) \tag{6}$$

$$(\lambda_1\lambda_2\lambda_3 = 1)$$

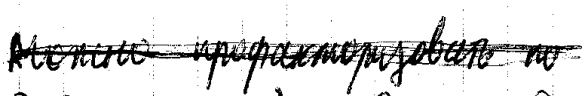
The space of homogeneous polynomials is finite-dimensional.

k_1, k_2, k_3 — the highest weight of the representation.

Formulae for the representation:

$$(T_g f)(z) = f(zg)$$

~~One can factor by~~.



Uncertain reading. "One can factor by" is crossed out.

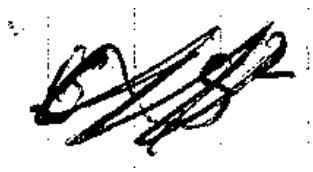
Original source details: GELFAND-P001503

The representation is specified by a triple of numbers (k_1, k_2, k_3).

Problem: 1) when is the representation trivial?

2) when is the representation tautological?

3) when is it [crossed out: in S^2] equivalent to the representation in S^2V?

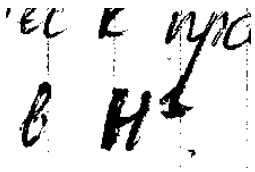


Uncertain reading. The crossed-out entry before "equivalent" appears to read "in S^2".

Original source details: GELFAND-P001503

Find the right basis in this space.

If the highest weight k is right, the representation is realized in sections of a bundle, i.e. in H^1.



Uncertain reading. The superscript of H resembles 1. A long stroke from the preceding line meets it.

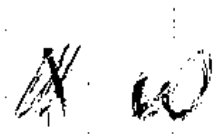
Original source details: GELFAND-P001503

Otherwise Bott discovered that the representation passes to higher cohomology of the flag space.

Gelf., Zelevinsky, Kapranov.

An important observation: on Z one must consider functions as well as differential forms!!!

Consider the [inserted: vector] field λ and all forms [crossed out] ω, for which $i_\lambda \omega = 0$.



Uncertain reading. The crossed-out sign before ω resembles λ.

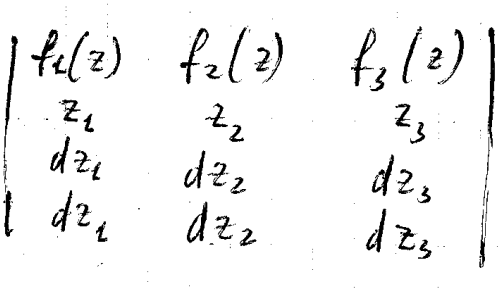
Original source details: GELFAND-P001503

Superalgebras of forms arise here.

Some important forms:

$$\Psi(z) \begin{vmatrix} z_1 & z_2 & z_3 \\ dz_1 & dz_2 & dz_3 \\ dz_1 & dz_2 & dz_3 \end{vmatrix}$$

$$\begin{vmatrix} f_1(z) & f_2(z) & f_3(z) \\ z_1 & z_2 & z_3 \\ dz_1 & dz_2 & dz_3 \\ dz_1 & dz_2 & dz_3 \end{vmatrix}$$



Uncertain reading. The final expression has four rows and three columns between vertical bars.

Original source details: GELFAND-P001503

GELFAND-P001504

[source number: 47]

All this is connected with the problem of what is meant by det.

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\det A \text{ — ?}$$

The denominator that arises when solving linear systems?

What if a_{ij} do not commute?

Another approach.

Vectors can be transformed by the action of the group GL_n.

There are 2 [crossed out] orbits: 0 and $V \setminus \{0\}$.



Undeciphered passage. The short entry between 2 and "orbits" is crossed out and illegible.

Original source details: GELFAND-P001504

Let X, Y be nonsingular matrices. One can replace A with XAY. Then all nonsingular matrices are equivalent.

det — a function $f(A)$, vanishing on singular matrices.

Or:

$$\sum a_{ik}\xi_i\eta_k$$



Uncertain reading. A sign resembling η is crossed out between ξ_i and η_k.

Original source details: GELFAND-P001504

X acts on ξ and Y on η.

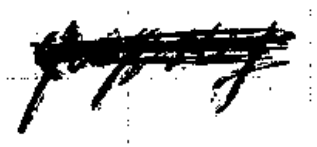
det — [inserted: homogeneous] function of the coefficients (a_{ij}) of least degree, equal to 0 on all singular surfaces.

(All this is on projective spaces).

Consider the cubic matrix a_{ijk} and [crossed out] the cone

$$\sum a_{ijk}\xi_i\eta_j\zeta_k = 0 \tag{7}$$

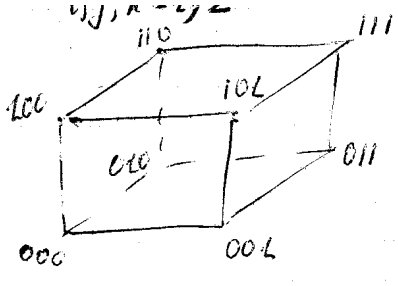
in projective space.



Uncertain reading. The word crossed out before "cone" may be "form".

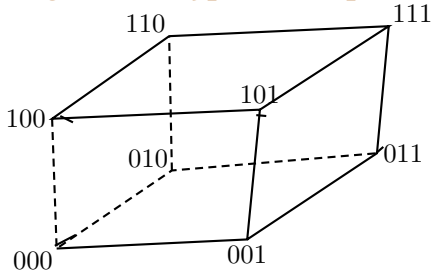
Original source details: GELFAND-P001504

Cubic det — a homogeneous function of least degree of a_{ijk}, equal to 0 on singular surfaces.
2-dimensional case: Cayley did it, but with an error
$i, j, k = 1, 2$



Original source details: GELFAND-P001504

Diagram and typed description.

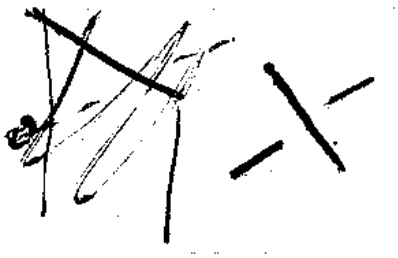


Several edges have gaps and short strokes. Some strokes resemble arrowheads.

deg $= 4$
Elements of 3 types: 1) squares (6 in all):

$$\sum \text{products over squares}$$

2) antisquares (inscribed tetrahedron)
(2 in all)



Original source details: GELFAND-P001504

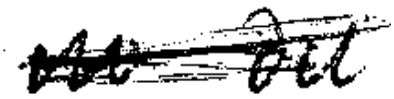
Diagram and typed description.



The left sketch has several overlapping strokes. In the right sketch one diagonal has a gap at the crossing.

GELFAND-P001505

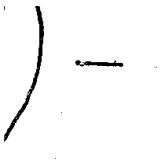
3) $\sum$ ~~over diag.~~ a^2b^2 (4 in all)
[beneath the sum: over the diagonals]



Original source details: GELFAND-P001505

Uncertain reading. “Over diag.” before a^2b^2 is crossed out.

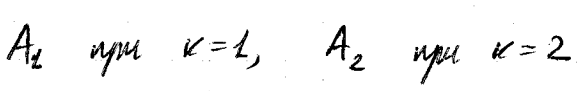
$$\mathrm{Det}(a_{ijk}) = \left(\begin{vmatrix} a_{111} & a_{212} \\ a_{121} & a_{222} \end{vmatrix} + \begin{vmatrix} a_{211} & a_{112} \\ a_{221} & a_{122} \end{vmatrix}\right)^2 - \\ - 4 \begin{vmatrix} a_{111} & a_{112} \\ a_{121} & a_{122} \end{vmatrix} \cdot \begin{vmatrix} a_{211} & a_{212} \\ a_{221} & a_{222} \end{vmatrix} \tag{8}$$



Uncertain reading. The minus sign is written twice, at the end of the first line and the beginning of the second.

Original source details: GELFAND-P001505

Consider 2 matrices A_1 at $k = 1$, A_2 at $k = 2$.



Uncertain reading. The line specifying the slices reads $k = 1$, $k = 2$.

Original source details: GELFAND-P001505

$\det |\xi_1 A_1 + \xi_2 A_2|$ — a quadratic form in ξ_1, ξ_2.

$$\mathrm{Discr.}\,[\det(\xi_1 A_1 + \xi_2 A_2)] = (8). \tag{9}$$

From (9) symmetry in all indices is not obvious.

(a_{ijk}) — product rep. $SL_n \times SL_n \times SL_n$

(on the three indices)

(tensor product of three tautological representations)

Hypothesis: det — Whitehead torsion of the Cayley–Koszul complex

(see p. 42)

Another approach

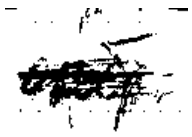
Consider the embedding

$$X = P^{m_1-1} \times P^{m_2-1} \times P^{m_3-1} \subset P^{m_1 m_2 m_3 - 1} = P$$

by Segre (given by the sheaf $O(1,1,1)$) — by multilinear forms.

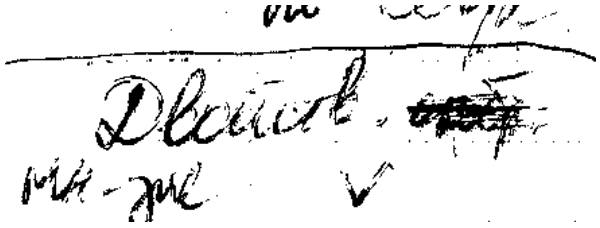
[At left, with a line to $\check{X}$: Dual [crossed out] set [abbrev.]]

Then $\check{X} \subset \check{P}$.



Undeciphered passage. An illegible word is crossed out after “Dual”.

Original source details: GELFAND-P001505



Uncertain reading. The abbreviation below “Dual” reads «мн-же» (set).

Original source details: GELFAND-P001505

$\check{X}$ — a hypersurface. deg — ? Equation — ?

$\check{P}$ — the space of hyperplanes in P.

$\check{X}$ — the set of hyperplanes tangent to X at some point.

[marked by a bracket:] tangency:

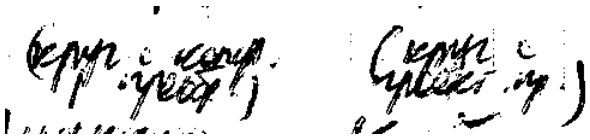
$$\pi \text{ is tangent to } X \;\Leftrightarrow\; T_x\pi \supset T_xX$$

One should read Klein

a) Hist. development of math. b) Higher geom. c) Elem. math. from the higher point of view.

Isomorphism of the Poincaré and Klein models for H^2?

[Above “Poincaré”: (disk with angles preserved). Above “Klein”: (disk with straight ...).]



Uncertain reading. The end of the note above “Klein” is unclear. It may read «yr.» (angles).

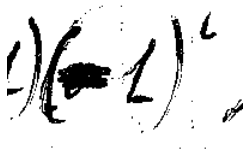
Original source details: GELFAND-P001505

GELFAND-P001506

[source number: 48]

$$N_d(2) = (-1)^d \sum_{i=0}^{d} (i+1)(-1)^i 2^{d-i} \binom{d}{i} i! \tag{10}$$

[In (10), the sign before 1 under the sum: minus?[?].]



Uncertain reading. minus?

Original source details: GELFAND-P001506

(degree of det in d-dimensional space)

[Inserted below:] of matrices $\underbrace{2 \times 2 \times \ldots \times 2}_{d \text{ times}}$.

Correct formula.

$$N_d(2) = (d+3)d!e_d(-2) + (d+1)(-2)^d \in \mathbb{Z}, \tag{11}$$

where

$$e_d(z) = \sum_{k=0}^{d-1} \frac{z^k}{k!}.$$

$$N_d(2) \sim \frac{(d+3)d!}{e^2} \quad \text{for large } d$$

d	$N_d(2)$	
2	2	
3	4	
4	24	(Schläfli)
5	128	

A. S. Schwarz.

Noncommutative analogue of Ray–Singer torsion.

Idea for constructing topological invariants in QFT

The action

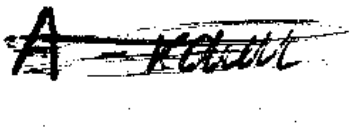
$$S(A) = \int_{M^3} A \wedge dA \tag{1}$$

[Crossed out after 1 in the equation number: 2?[?]. Crossed out at right: A — gau...[?].]



Uncertain reading. 2?

Original source details: GELFAND-P001506



Uncertain reading. A — gau...

Original source details: GELFAND-P001506

gives Ray–Singer torsion.

There is a generalization when A is a form with values in the Lie algebra $\mathfrak{g}$

(gauge field or connection in a principal bundle (trivial))

$M^3 \times G$

G is a compact Lie group

$$S(A) = \operatorname{Tr} \int_{M^3} A \wedge dA + \frac{2}{3} \operatorname{Tr} \int_{M^3} A \wedge A \wedge A \tag{2}$$

[The lettering in the second term has been overwritten: Tr?[?].]



Uncertain reading. Tr?

Original source details: GELFAND-P001506

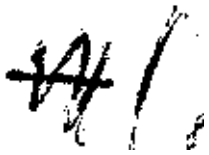
[Below the second term:] $\operatorname{Tr}[A, A] \wedge A$

one should consider the partition function

$$Z = \int e^{-S(A)} \tag{3}$$

$$\delta S = 0 \quad \text{(equation of motion)} \quad \Leftrightarrow \quad F = 0 \tag{4}$$

[Crossed out after $\delta S = 0$: yp?[?].]



Uncertain reading. yp?

Original source details: GELFAND-P001506

[Below F:] curvature of the connection.

GELFAND-P001507

One can write

$$S(A) = \int_{N^4} F \wedge F, \qquad \partial N = M \tag{5}$$

(extend the connection A to N^4 and take the curvature F)

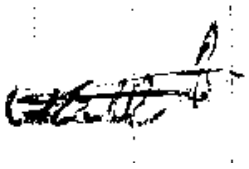
Here unresolved word arise [wavy underline].



Uncertain reading. unresolved word

Original source details: GELFAND-P001507

Link $\rightsquigarrow$ neighbourhood of the knot, [crossed out: glu...?] glue it to the double — — obtain M^3.



Uncertain reading. glu...?

Original source details: GELFAND-P001507

Also consider [Crossed out before P: Tr?.]



Uncertain reading. Tr?

Original source details: GELFAND-P001507

$$P \exp \oint_\Gamma A\,dx \quad \text{(monodromy)}$$

and modify

$$S(A) = \int e^{-S(A)} \operatorname{Tr} P \exp \oint_\Gamma A\,dx \tag{6}$$

[An opening parenthesis is in the margin before (6). Left-hand side of (6): $S(A)$?.]



Uncertain reading. $S(A)$

Original source details: GELFAND-P001507

GELFAND-P001508

[source number: 53]

31/X-88. Gelfand Seminar.

V. E. Zakharov. Equations of hydrodynamics in Clebsch form.

Let S be a matrix from a semisimple Lie algebra. Consider the equation

$$\frac{\partial S}{\partial t} = \left[S, \frac{\partial H}{\partial S}\right] = S_{\alpha\gamma}\frac{\partial H}{\partial S_{\gamma\beta}} - \frac{\partial H}{\partial S_{\alpha\gamma}}S_{\gamma\beta} \tag{1}$$

[Crossed out after the second equals sign: H?.]

Uncertain reading. H?



Original source details: GELFAND-P001508

$$\mathrm{Sp}\, S^n = I_n \quad \text{— integral of motion} \tag{2}$$

$$S = [p, q], \qquad H = H(S) = H(p, q] \qquad (p, q \text{ are letters}) \tag{3}$$

[The closing delimiter in $H(p, q]$: square bracket or sharply curved parenthesis[?].]



Uncertain reading. square bracket or sharply curved parenthesis

Original source details: GELFAND-P001508

$$\begin{cases} \dot{p} = \left[p, \dfrac{\partial H}{\partial S}\right] = -\dfrac{\partial H}{\partial q}, \\ \dot{q} = \left[q, \dfrac{\partial H}{\partial S}\right] = \dfrac{\partial H}{\partial p}. \end{cases} \tag{4}$$

$$\mathrm{Sp}(p^{n_1} q^{n_2} p^{n_3} \ldots) = \text{const} \quad \text{— integrals} \tag{5}$$

Clebsch variables

$\varphi(x, y)$

$$A = -\varphi_y \frac{\partial}{\partial x} + \varphi_x \frac{\partial}{\partial y} \tag{6}$$

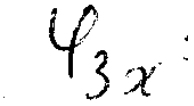
Suppose there are 2 such A_1, A_2.

Then

$$[A_1, A_2] = -\varphi_{3y} \frac{\partial}{\partial x} + \varphi_{3x} \frac{\partial}{\partial y} \tag{7}$$

$$\varphi_{3x} = \varphi_{1x}\varphi_{2y} - \varphi_{1y}\varphi_{2x} = \{\varphi_1, \varphi_2\}$$

[Left-hand side of the bracket definition: φ_{3x}[?].]



Uncertain reading. φ_{3x}

Original source details: GELFAND-P001508

Equations of hydrodynamics in Clebsch form

$$\frac{\partial \Omega}{\partial t} + \{\Psi, \Omega\} = 0 \tag{8}$$

[Crossed out at right: illegible[?].]



Uncertain reading. illegible

Original source details: GELFAND-P001508

$v_x, v_y, \quad \operatorname{div} v = 0$

$$v_x = -\Psi_y, \quad v_y = \Psi_x \qquad (\Psi \text{ is the stream function}) \tag{9}$$

$$\Omega = \Delta\Psi \tag{10?}$$

[Overwritten equation number: 10?[?].]

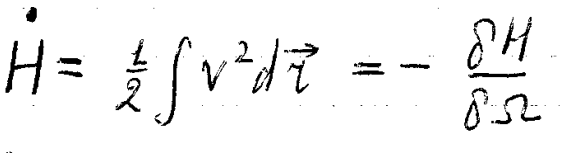


Uncertain reading. 10?

Original source details: GELFAND-P001508

$$\dot{H} = \frac{1}{2}\int v^2\, d\vec{r} = -\frac{\delta H}{\delta\Omega} \tag{11}$$

[Line with a dot above H and two equals signs: $\dot{H} = \frac{1}{2}\int v^2\, d\vec{r} = -\frac{\delta H}{\delta\Omega}$?.]

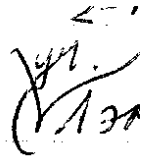


Uncertain reading. $\dot{H} = \frac{1}{2}\int v^2\, d\vec{r} = -\frac{\delta H}{\delta\Omega}$

Original source details: GELFAND-P001508

(Hamiltonian formulation of 2-dimensional hydrodynamics)

Known since 1844 [inserted below: (Clebsch)] ([written above the parenthesis: eqs.[?]] of Lamb et al.)



Uncertain reading. eqs.

Original source details: GELFAND-P001508

Let $\Omega = \{\lambda, \mu\}$. One gets, as in (4):

[Below $\{\lambda, \mu\}$:] (Poisson bracket)

One gets:

$$\begin{cases} \dfrac{\partial\lambda}{\partial t} + \{\Psi, \lambda\} = 0 \\ \dfrac{\partial\mu}{\partial t} + \{\Psi, \mu\} = 0 \end{cases} \quad \Bigg| \quad \begin{cases} \dfrac{\partial\lambda}{\partial t} = -\dfrac{\delta H}{\delta\mu} \\ \dfrac{\partial\mu}{\partial t} = \dfrac{\delta H}{\delta\lambda} \end{cases} \tag{12}$$

Integrals of motion?

We have

$$\Psi = -\frac{\delta H}{\delta\Omega} \tag{13}$$

and

$$\frac{\partial\Omega}{\partial t} + \{\Psi, \Omega\} = 0 \tag{14}$$

(15) Lemma 1 If

$$\frac{\partial\lambda_i}{\partial t} + \{\Psi, \lambda_i\} = 0, \qquad i = 1, 2, \tag{16}$$

then $\{\lambda_1, \lambda_2\}$ satisfies the same equation

$$\frac{\partial}{\partial t}\{\lambda_1, \lambda_2\} + \{\Psi, \{\lambda_1, \lambda_2\}\} = 0. \tag{17}$$

Lemma 2.

If λ_i, $i = 1, \ldots, N$ satisfy (16), then

$$\int F(\lambda_1, \ldots, \lambda_N)\, d\vec{r} = \text{const} \quad \text{— integral of motion} \tag{18}$$

Thus the following are conserved

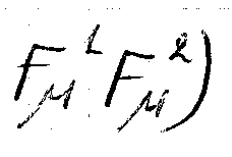
$$\int F(\lambda, \mu, \Omega, \{\lambda, \Omega\}, \{\mu, \Omega\}, \ldots)\, d\vec{r} \tag{19}$$

The Lie algebra of functionals (19) (with respect to the “Poisson bracket”) contains the subalgebra (!)

$$J_0 = \left\{ \int F(\lambda, \mu)\, d\vec{r} \right\} \tag{20}$$

$$\begin{aligned} [J_0^1, J_0^2] &= \int (F_\lambda^1 F_\mu^2 - F_\mu^1 F_\mu^2)\, d\vec{r} \\ &= \int \langle F_1, F_2 \rangle\, d\vec{r} \end{aligned} \tag{21}$$

[Last factor in parentheses in (21): $F_\mu^{2?}$.]



Uncertain reading. F_μ^2

Original source details: GELFAND-P001509

[Below $\langle F_1, F_2 \rangle$:] ordinary Poisson bracket on phase space.

Consider the algebra

$$J_1 = \left\{ \int F(\lambda, \mu, \Omega)\, d\vec{r} \right\} \tag{22}$$

This will be a current algebra over the algebra (20).

GELFAND-P001510

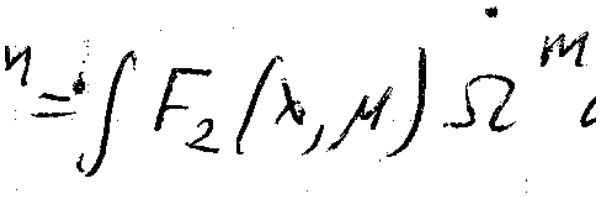
[source number: 54]

Consider

$$J_1^n = \int F_1(\lambda,\mu)\Omega^n\, d\vec{r}, \qquad J_2^m = \int F_2(\lambda,\mu)\Omega^m\, d\vec{r} \tag{23}$$

[In the right-hand formula in (23) there are dots near the equals sign and above Ω: their significance is unclear?.]

Uncertain reading. their significance is unclear

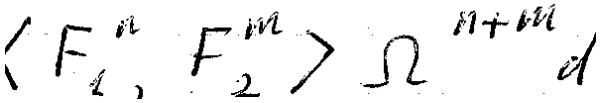


Original source details: GELFAND-P001510

$$[J_1^n, J_2^m] = \frac{(n-1)(m-1)}{n+m-1}\int \langle F_1^n, F_2^m\rangle \Omega^{n+m}\, d\vec{r} \tag{24}$$

[In (24): $\langle F_1^n, F_2^m\rangle\Omega^{n+m}$?.]

Uncertain reading. $\langle F_1^n, F_2^m\rangle\Omega^{n+m}$



Original source details: GELFAND-P001510

(one obtains the Virasoro algebra with zero central charge)

There is a 2-parameter family of integrals. There is no integrability in the general case (H is arbitrary!!) What kind of incompleteness?

Let us try applying Liouville's theorem.

We make changes of variables.

$$\begin{cases} x' = x'(x,y) \\ y' = y'(x,y) \end{cases} \qquad \frac{\partial(x',y')}{\partial(x,y)} \equiv 1. \tag{25}$$

preserve area. All the integrals (23) are unchanged by this.

Fix λ. Which changes preserve λ

One has to slide along the level curves of λ. $\varphi = f(\lambda)$.

Introduce

$$a_k = \frac{1}{\sqrt{2}}(\lambda_k + i\mu_k)$$

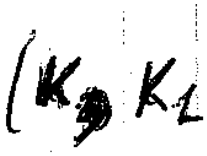
and consider the Hamiltonian

$$H = \int W_{kk_1,k_2k_3}\bar{a}_k\bar{a}_{k_1}a_{k_2}a_{k_3} \tag{26}$$

For its special form (the exact expression is left unwritten here)

$$W'(k,k_1,k_2,k_3) = \{k_1,k_2\}\{k_2,k_3\}f(k_1-k_3)+\dots \tag{27}$$

[After the first k in (27): overwritten comma or subscript followed by a comma?.]



Uncertain reading. overwritten comma or subscript followed by a comma

Original source details: GELFAND-P001510

$(28)_1$ [Crossed out: For] When $f = \frac{1}{\vec{k}^2}$, there are too few integrals.

$(28)_2$ When $f = \frac{\vec{\alpha}\,\vec{k}}{\vec{\beta}\,\vec{k}}$, one obtains an ∞-dimensional top and it is integrable.

Zakharov wants to conduct a numerical experiment on the onset of chaos.

[Inserted at left:] In the 3-dimensional case

$$\vec{\Omega} = [\vec{A}, \operatorname{rot}\vec{B}] - [\vec{B}, \operatorname{rot}\vec{A}]$$

Now a vector, and integrals do not appear in sufficient number.

At the beginning of the next seminar Zakharov will speak about turbulence.

GELFAND-P001511

I. M. Gelfand.

[Crossed out and hatched below the name: illegible?.]



Uncertain reading. illegible

Original source details: GELFAND-P001511

Dual curve

$$y = kx + b$$



Diagram and typed description.



Original source details: GELFAND-P001511

this is inconvenient because k and b have different dimensions.

$$\frac{x}{a}+\frac{y}{b}=1$$ also less than satisfactory.

Better:

$$\xi x+\eta y=1$$



Original source details: GELFAND-P001511

Diagram and typed description.



A straight line is a point in the ξ,η plane

Similarly:

$$\xi_1 x_1+\ldots+\xi_n x_n=1$$ — a hyperplane.

Then

the dual curve is formed by all the tangents in the (ξ,η) plane

Example 1. $y=x^3$

$$y-y_0=3x_0^2(x-x_0)$$
$$k=3x_0^2,\quad b=-2x_0^3$$



Original source details: GELFAND-P001511

Diagram and typed description.



dual curve

From this one can understand the Legendre transform.

Example 2.

$$|x|^p+|y|^p=1$$

It is best to take

$$\xi x+\eta y=1$$ and find when it is tangent.

One obtains

$$|\xi|^q+|\eta|^q=1,\qquad \text{where}\quad \frac{1}{p}+\frac{1}{q}=1$$

[The first exponent has been overwritten: q; the earlier mark is illegible?.]



Original source details: GELFAND-P001511

Uncertain reading. q; the earlier mark is illegible

(dual Banach space)

Problem (for enthusiasts of the geometry of Banach spaces)

A sufficiently high-dimensional cube has a section arbitrarily close to a circle.

Complex point of view.

Dual set — tangent planes

GELFAND-P001512

[source number: 55]

In $\mathbb{C}^3$:

$$\underline{\text{line}} \quad \longleftrightarrow \quad \underline{\text{line}}$$

The dimension of the dual set is an important problem.

Problem Consider the space

$$\mathbb{P}^{(m_1-1)} \times \mathbb{P}^{(m_2-1)} \times \ldots \times \mathbb{P}^{(m_k-1)} \subset \mathbb{P}^{m_1 m_2 \ldots m_k - 1}$$

[Crossed out: Problem.] The problem of hyperdeterminants is equivalent to the following problem:

when does the dual surface have codim $= 1$?

The left-hand side of the equation defining the dual surface is the hyperdeterminant.

Degree — ?

Gelfand and coauthors can compute this. The answer is as follows: for

$$\mathbb{P}^{\ell_1} \times \mathbb{P}^{\ell_2} \times \ldots \times \mathbb{P}^{\ell_d} \subset \mathbb{P}^{(\ell_1+1)(\ell_2+1)\ldots(\ell_d+1)-1}$$

the degree of the polynomial defining [crossed out: ϕ?[?]] it will be

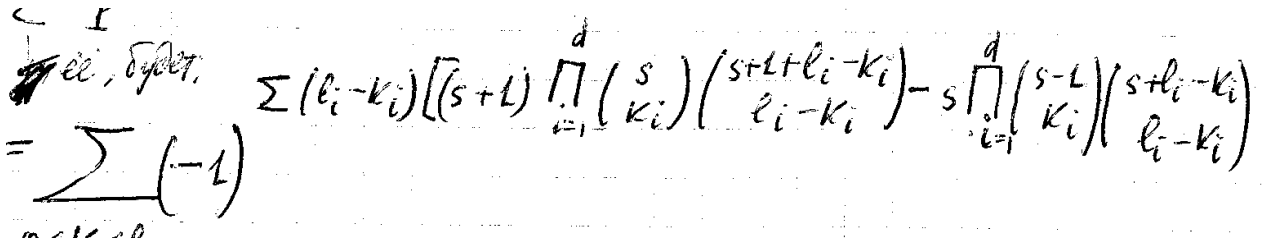
Uncertain reading. ϕ?



Original source details: GELFAND-P001512

$$N(\ell_1, \ell_2, \ldots, \ell_d) = \sum_{\substack{0 \le k_1 \le \ell_1 \\ \ldots \\ 0 \le k_d \le \ell_d}} (-1)^{\sum(\ell_i - k_i)} \Big[(s+1) \prod_{i=1}^{d} \binom{s}{k_i} \binom{s+1+\ell_i-k_i}{\ell_i - k_i} - s \prod_{i=1}^{d} \binom{s-1}{k_i} \binom{s+\ell_i-k_i}{\ell_i-k_i} \tag{1}$$

[In (1), the continuation after (-1) is raised to the level of the exponent and the closing square bracket is not visible: the expression after the sum in the exponent is typeset as a multiplier[?].]



Original source details: GELFAND-P001512

Uncertain reading. the expression after the sum in the exponent is typeset as a multiplier

$$s = k_1 + \ldots + k_d$$

D. Leshchiner.

Generating function for this number

$$\begin{aligned} &\sum N(\ell_1, \ldots, \ell_d, \ldots)\, x_1^{\ell_1} x_2^{\ell_2} \ldots x_d^{\ell_d} x_{d+1}^0 \ldots \\ &= \left(\sum_{k=0}^{\infty} (1-k) e_k \right)^{-2} \end{aligned} \tag{2}$$

[The first variable in (2) has been overwritten: x_1 or z_1?.]



Uncertain reading. x_1 or z_1

Original source details: GELFAND-P001512

[Crossed out after $x_2^{\ell_2}$: ℓ?.]



Uncertain reading. ℓ?

Original source details: GELFAND-P001512

(after ℓ_d we append zeros successively?)



Uncertain reading. successively?

Original source details: GELFAND-P001512

$$e_0 = 1$$

e_k are the elementary symmetric functions:

$$\begin{aligned} e_1 &= \sum_i x_i, \\ e_2 &= \sum_{i<j} x_i x_j \\ e_3 &= \sum_{i<j<k} x_i x_j x_k \\ &\ldots \end{aligned}$$

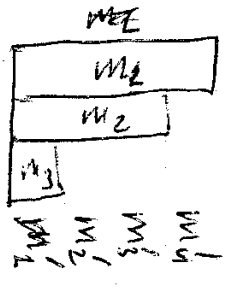
M. Kapranov. Another formula:

$$\lambda = (\ell_1 \geq \ell_2 \geq \ldots)$$

$$N(\lambda) = \sum_{\mu} \frac{(m_1+1)! \prod_{i \geq 1} (i-1)^{m_i - m_{i+1}}}{(m_1 - m_2)!\,(m_2 - m_3)! \ldots} \cdot M(\mu', \lambda),$$

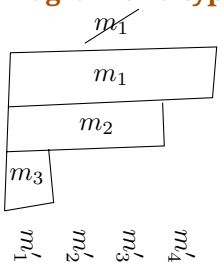
$$\mu = (m_1 \geq m_2 \geq \ldots)$$

where μ' is the conjugate partition



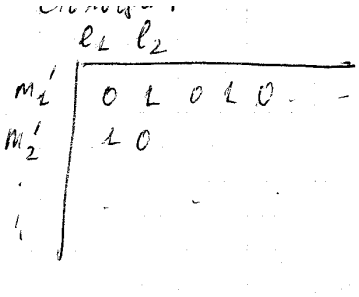
Original source details: GELFAND-P001513

Diagram and typed description.



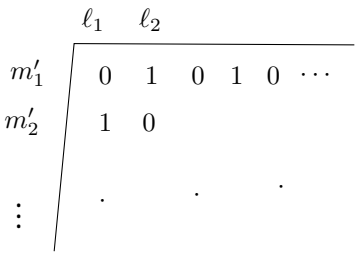
[Above the top strip a label resembling m_1 is crossed out. Below the diagram the first rotated label, m'_1, is overwritten.]

$M(\mu', \lambda)$ is the number of matrices of 0 and 1 whose row and column sums



Original source details: GELFAND-P001513

Diagram and typed description.



$$M(\mu', \lambda) > 0 \quad \Rightarrow \quad \lambda \geq \mu \quad \text{with respect to the natural order.}$$

and, moreover, then

$$\forall i: \quad \ell_i \leq \sum_{j \neq i} \ell_j$$

Zelevinsky

Boundary case:

$$\ell_1 = \ell_2 + \ldots + \ell_d$$

Then

$$N = \frac{(\ell_1 + 1)!}{\ell_2! \ell_3! \ldots \ell_d!}$$

Apparently this is where one should compute det

(1st nontrivial case: [crossed out: illegible mark?] $2 \times 3 \times 4$)

Uncertain reading. illegible mark



Original source details: GELFAND-P001513

GELFAND-P001514

13/II-88.

Zelevinsky. On multidimensional determinants.

$$A = (a_{i_1 \ldots i_d})_{0 \le i_k \le \ell_k} \quad \text{— a multidimensional matrix}$$

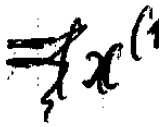
$$\dim V_k = \ell_k + 1, \qquad x_i^{(1)} \in V_1, \ldots, x_i^{(d)} \in V_d.$$

$$A \in (V_1 \otimes \ldots \otimes V_d)^*$$

$$x = x^{(1)} \otimes \ldots \otimes x^{(d)}$$

[A stroke crosses the two bars before $x^{(1)}$: $=$ or $\neq$?.]

Uncertain reading. $=$ or $\neq$

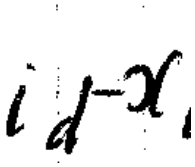


Original source details: GELFAND-P001514

$$A(x) = \sum_{i_1, \ldots, i_d} a_{i_1 \ldots i_d} - x_{i_1}^{(1)} \ldots x_{i_d}^{(d)}$$

[After $a_{i_1 \ldots i_d}$ there is a short stroke: minus or linking stroke?.]

Uncertain reading. minus or linking stroke



Original source details: GELFAND-P001514

$$\text{Det } A = 0 \iff A \text{ is degenerate}$$

$$A \text{ is degenerate} \overset{\text{def}}{\iff} \exists x^{(1)}, \ldots, x^{(d)} \neq 0$$

$$\ell_1 \ge \ell_2, \ell_3, \ldots, \ell_d$$

$$A(x^{(1)}, \ldots, x^{(i-1)}, \cdot, x^{(i+1)}, \ldots, x^{(d)}) \equiv 0 \quad \text{for all } i$$

$$\text{Det is defined} \overset{\text{def}}{\iff} \text{Codim}\{\text{degenerate matrices}\} = 1$$

$$\Updownarrow$$

$$\ell_1 \le \ell_2 + \ell_3 + \ldots + \ell_d$$

$$(\ell_1 + 1) \times (\ell_1 + 1) \quad \Rightarrow \quad \deg = \ell_1 + 1$$

(1) Claim. Suppose $\ell_1 = \ell_2 + \ldots + \ell_d$

Then Det has degree

$$\frac{(\ell_1 + 1)!}{\ell_2! \ell_3! \ldots \ell_d!}$$

Let the degree of Det for $(\ell_1 + 1) \times \ldots \times (\ell_d + 1)$ be $N(\ell_1, \ldots, \ell_d)$

and $N(\ell_1, \ldots, \ell_d) = 0$ if Det is undefined

GELFAND-P001515

[source number: 69]

(2) Claim.

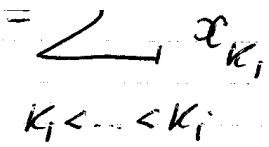
$$\sum_{\ell_1,\ldots,\ell_d\geq 0} N(\ell_1,\ldots,\ell_d)x_1^{\ell_1}\ldots x_d^{\ell_d} =$$

$$= \left[1-\sum_{i\geq 2}(i-1)e_i(x_1,\ldots,x_d)\right]^{-2},$$

where e_i is the elementary symmetric polynomial

$$e_i(x_1,\ldots,x_d) = \sum_{k_i<\ldots<k_i} x_{k_i}\ldots x_{k_i}$$

[The first lower index in the summation condition and in the first factor: i or j?.]



Uncertain reading. i or j

Original source details: GELFAND-P001515

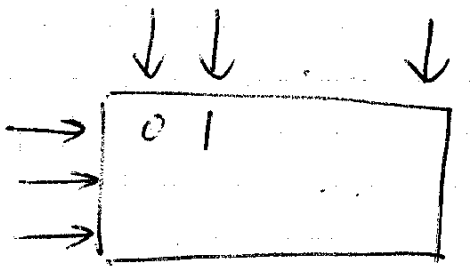
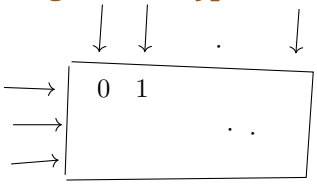


Diagram and typed description.



Original source details: GELFAND-P001515

Consider $0,1$-matrices with given row and column sums.

Everything is expressed in terms of such numbers.

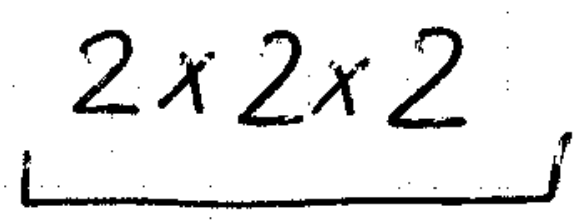


Diagram and typed description.

$2\times 2\times 2$

Original source details: GELFAND-P001515

A complex will be constructed

$$0 \longrightarrow C_0 \xrightarrow{d(A)} C_1 \xrightarrow{d(A)} \dots \xrightarrow{d(A)} C_n \longrightarrow 0 \tag{3}$$

[In the right margin:] ‖ $d(A)$ depends linearly on the matrix A

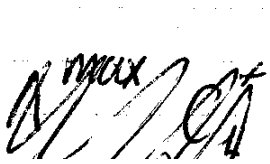
A is degenerate $\iff$ failure of exactness

Let $0 \to C_0 \to C_1 \to \dots \to C_n \to 0$

$$\operatorname{Det} C_\bullet = \frac{\Lambda^{\max} C_0 \otimes \Lambda^{\max} C_2 \otimes \dots}{\Lambda^{\max} C_1 \otimes \Lambda^{\max} C_3 \otimes \dots}$$

[Crossed out: $\Lambda^{\max} C_1$ or $\Lambda^{\max} C_i$?.]

Uncertain reading. $\Lambda^{\max} C_1$ or $\Lambda^{\max} C_i$



Original source details: GELFAND-P001515

$$\sum (-1)^i \dim C_i = \sum (-1)^i \dim H_i$$

A natural isomorphism follows from the proof.

$$\operatorname{Det} C_\bullet \cong \operatorname{Det} H_\bullet$$

Acyclicity of $C_\bullet \Rightarrow \operatorname{Det} H_\bullet = \mathbb{C}$.

GELFAND-P001516

Choose bases in all the $C_i \Rightarrow$ this gives an element [crossed out: in?]

Uncertain reading. in?



Original source details: GELFAND-P001516

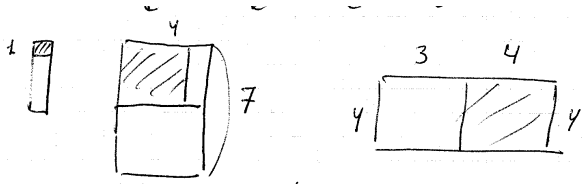
$$\alpha \in \operatorname{Det} C_\bullet$$

and, consequently, a number.

α depends on A rationally and often polynomially.

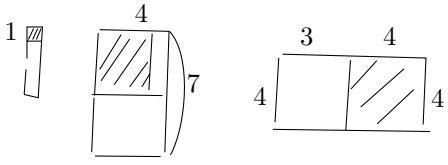
For the 2×2 matrix determinant $ad - bc$, one obtains

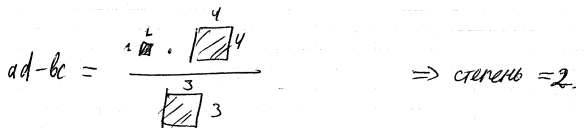
$$0 \to \mathbb{C} \to \mathbb{C}^4 \to \mathbb{C}^7 \to \mathbb{C}^4 \to 0$$



Original source details: GELFAND-P001516

Diagram and typed description.





Original source details: GELFAND-P001516

Diagram and typed description.

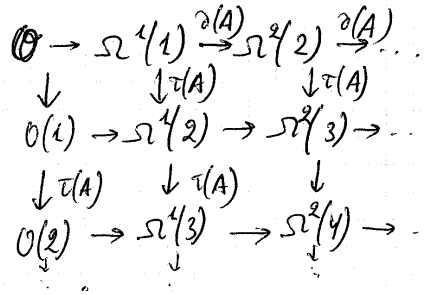
$$ad - bc = \frac{1\frac{1}{4} \cdot 4}{3} \Rightarrow \text{degree} = 2.$$

det = GCD of the maximal minors of the 4×7 matrix

— · —

Kapranov

$$X = P(V_1^*) \times \cdots \times P(V_d^*), \qquad O(1) = O(1, \ldots, 1)$$



Original source details: GELFAND-P001516

Diagram and typed description.

$$\begin{array}{ccccccc}
O & \longrightarrow & \Omega^1(1) & \xrightarrow{\partial(A)} & \Omega^2(2) & \xrightarrow{\partial(A)} & \cdots \\
\downarrow & & \downarrow{\scriptstyle \tau(A)} & & \downarrow{\scriptstyle \tau(A)} & & \\
O(1) & \longrightarrow & \Omega^1(2) & \longrightarrow & \Omega^2(3) & \longrightarrow & \cdots \\
\downarrow{\scriptstyle \tau(A)} & & \downarrow{\scriptstyle \tau(A)} & & \downarrow & & \\
O(2) & \longrightarrow & \Omega^1(3) & \longrightarrow & \Omega^2(4) & \longrightarrow & \cdots \\
\downarrow & & \downarrow & & \downarrow & &
\end{array}$$

[Upper-left node: O beneath the overdrawn strokes?[?].]



Uncertain reading. O beneath the overdrawn strokes?

Original source details: GELFAND-P001516

[At the upper end of the arrow $\Omega^1(1) \to \Omega^1(2)$: a slanted stroke resembling part of a second arrowhead[?].]



Uncertain reading. a slanted stroke resembling part of a second arrowhead

Original source details: GELFAND-P001516

1) $\partial(A)^2 = 0, \qquad \partial(A)\partial(B) = -\partial(B)\partial(A)$
2) $\tau(A)\partial(A) = \partial(A)\tau(A)$
3) $\tau(A)\tau(B) = \tau(B)\tau(A)$

GELFAND-P001517

[source number: 70]

Then fixing $m \geq 0$, we obtain

$$C_i^{(m)} \cong \Gamma(X, \Omega^i(i+m)) \oplus \Gamma(X, \Omega^{i-1}(i+m))$$

[Sign between the left- and right-hand sides: $\cong^?$.]

Uncertain reading. $\cong$



Original source details: GELFAND-P001517

[Below the formula, at right:] (totalization of a bicomplex consisting of 2 rows)

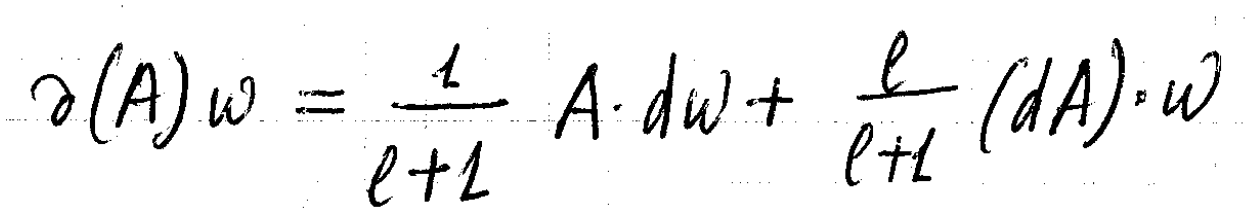
$$A \in \Gamma(X, O(1))$$

$\tau(A)$ is multiplication by $A \in \Gamma(X, O(1))$

$\partial(A)$ on $\Omega^k(\ell)$

$$\partial(A)\omega = \frac{1}{\ell+1} A \cdot d\omega + \frac{\ell}{\ell+1}(dA) \cdot \omega$$

[The manuscript reads: a $+$ between the terms and coefficients $\frac{1}{\ell+1}$ and $\frac{\ell}{\ell+1}$?.]



Original source details: GELFAND-P001517

Uncertain reading. a $+$ between the terms and coefficients $\frac{1}{\ell+1}$ and $\frac{\ell}{\ell+1}$

— · —

Zelevinsky

$$X \subset \mathbb{P}(V^*)$$

[Above the abbreviation for “linearly”: 1 or a vertical stroke?.]

Uncertain reading. 1 or a vertical stroke



Original source details: GELFAND-P001517

linearly normal, i.e. $H^0(X, O(1)) =$ [crossed out: illegible mark?] V.

Uncertain reading. illegible mark



Original source details: GELFAND-P001517

Examples

1) V is an irreducible representation of a semisimple group G

$X \subset \mathbb{P}(V^*)$ is the projectivization of the orbit of the highest-weight vector.

2) Let a complex[?] torus $(\mathbb{C}^*)^n = H$ act on V

Uncertain reading. complex



Original source details: GELFAND-P001517

Characters $\mathbb{Z}^n$, with a distinguished collection in it

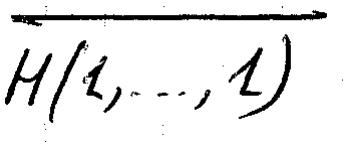
$$\mathbb{Z}^n = ((\mathbb{C}^*)^n)^\wedge \ni \chi_1, \dots, \chi_N, \qquad N = \dim V, \quad V = \mathbb{C}^N$$

All the χ_j lie on an affine hyperplane in $\mathbb{Z}^n$

$$\begin{aligned} X &= (\text{closure of a generic orbit of this complex torus}) \\ &= \overline{H(1, \dots, 1)} \end{aligned}$$

[Ends of the bar above $H(1, \dots, 1)$: strokes resembling arrowheads[?].]

Uncertain reading. strokes resembling arrowheads



Original source details: GELFAND-P001517

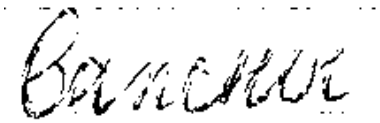
Intersection of examples 1 and 2 — hyperdeterminants.

GELFAND-P001518

[source number: 73]

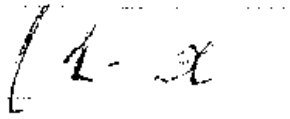
The behavior of the Betti number on the space of representations has been little studied! Jumps in the Betti number are important[?] (first?[?] triv. unresolved word!)

Uncertain reading. important



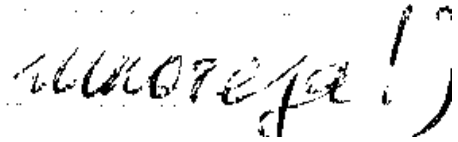
Original source details: GELFAND-P001518

Uncertain reading. first?



Original source details: GELFAND-P001518

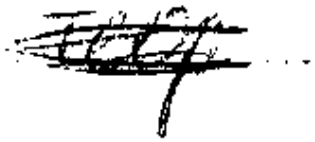
Uncertain reading. unresolved word



Original source details: GELFAND-P001518

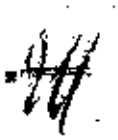
Conjecture: [Crossed out: illegible[?].] The Jones polynomial is merely [crossed out: illegible[?].] a scheme for computing the invariants that have to be computed first.

Uncertain reading. illegible



Original source details: GELFAND-P001518

Uncertain reading. illegible



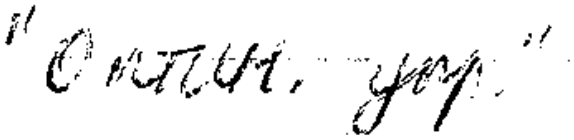
Original source details: GELFAND-P001518

20/II-89 Plan for Vishik's seminar

[Left column:]

1. Fursikov "Optimal control[?]"
2. Vainberg
3. Kuksin[?].
4. Agranovich

Uncertain reading. Optimal control



Original source details: GELFAND-P001518

Uncertain reading. Kuksin



Original source details: GELFAND-P001518

[Crossed out below item 4: illegible[?].]

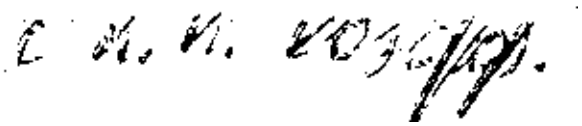
Uncertain reading. illegible



Original source details: GELFAND-P001518

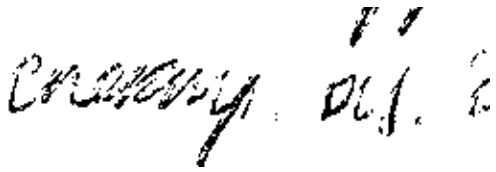
[Right column, beyond the vertical rule:]

5. Sinai “Schrödinger operator with q.p.[?] coefficients”
6. Birman or Solomyak (spectral asymptotics[?] for the Schrödinger operator)
7. Babin, Vishik: on attractors for unbounded domains.
8. Tuzhilin “Minimal surfaces”
9. Gelfand, Zakharevich.



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Uncertain reading. q.p.



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Uncertain reading. spectral asymptotics

20/II-89 Beilinson.
Lusztig.
The following objects are roughly equivalent

1. Representations of [crossed out: illegible[?]] semisimple algebraic groups in finite characteristic.
2. Representations of quantum groups when $q =$ a root of 1.
3. Representations of loop groups with integral level.
4. Geometry of affine Schubert cells



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Uncertain reading. illegible



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Diagram and typed description.

highest wt. $0\ 1\ 2$ ℓ' $p-1$ ℓ $2p-1$

dim. 1 p $2p$

[An illegible mark is crossed out below 0. The arrowheads on the arc are faint.]

X_ℓ is the irreducible representation of highest weight ℓ

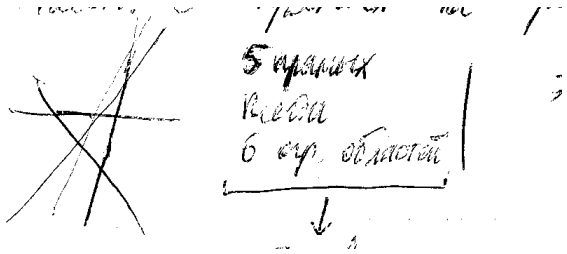
V_ℓ is the representation on homogeneous polynomials.

Let ℓ be an integer, then

$$0 \to X_{\ell'} \to V_\ell \to X_\ell \to 0$$

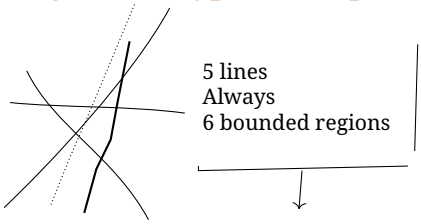
I. M. Gelfand.

Consider 6 lines in the projective plane



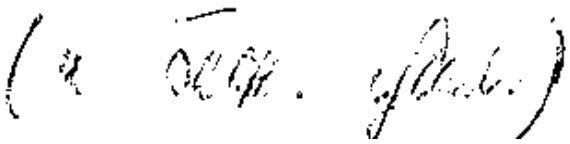
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Diagram and typed description.



[The thin steep line is visible in places. Some intersections are indistinct.]

[Below the figure:] (and[?] the line at infinity)



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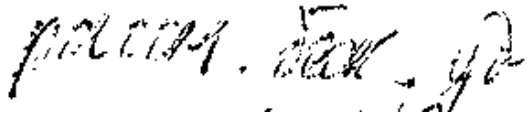
Uncertain reading. and

[Further right:] This occurs in the integral

$$\int \frac{dx}{\sqrt{(x-a)(x-b)(x-c)(x-d)}}$$

[Below the arrow:] Proof: consider the infinitely distant[?] line.

Uncertain reading. infinitely distant



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GELFAND-P001519

Kapranov

X is a $K3$ surface if

1) $\pi_1(X) = 0$

2) Ω^2_X is trivial (as a holomorphic bundle)

[Written below Ω^2_X:] (bundle of volume forms)
(i.e. $\exists$ a holomorphic volume form [crossed out: illegible?] which is nowhere zero)

Uncertain reading. illegible



Original source details: GELFAND-P001519

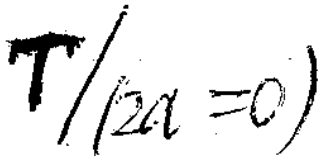
They are all homeomorphic and even diffeomorphic.

Example: 2-dimensional complex torus

$$\begin{array}{c} T = \mathbb{C}^2/\mathbb{Z}^4 \\ \downarrow \qquad\qquad (x \text{ and } -x \text{ identified}) \\ T/(2x = 0) \end{array}$$

[Notation for the quotient: $T/(2x = 0)$?.]

Uncertain reading. $T/(2x = 0)$



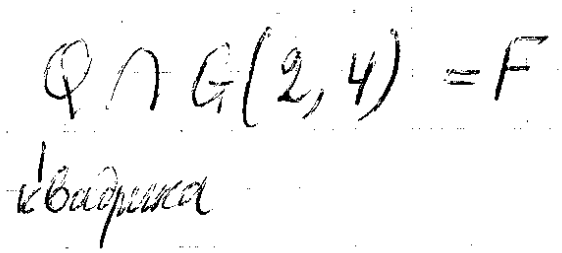
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16 singular points arise (Kummer surface)
It will be smooth after resolving the singularities.

— · —

$G(2,4) \subset \mathbb{P}^5$ (quadric)

Diagram and typed description.
$Q \cap G(2,4) = F$
|
quadric



Original source details: GELFAND-P001519